

# Module 2: Comparing Groups

## Simple Statistical Tests in Python

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# What We Will Do in This Module

- 1 Understand what a p-value is
- 2 Compare two groups (t-test)
- 3 Compare two paired measurements (paired t-test)
- 4 Compare three or more groups (ANOVA)
- 5 Check whether two variables move together (correlation)
- 6 Predict one variable from another (regression)

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These are called **inferential tests**. They help us decide whether a difference in the data is likely to be real or just a coincidence.

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A test is like asking: "If there were really no difference, how often would we see a difference this large just by luck?"

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Think of 0.05 as a common threshold. Below it, we say the difference is **statistically significant**. Above it, we say we cannot be confident.

## Before We Begin: Load the Data

```
import pandas as pd
from scipy import stats

df = pd.read_csv("village_survey.csv")

print("Data loaded.")
```

```
Data loaded.
```

We reuse the same dataset from Module 1. The new tool here is `scipy`, which runs statistical tests.

# Test 1: Comparing Two Groups

- We want to know: is urban income different from rural income?
- We have two groups: Rural and Urban
- Each household belongs to one group
- We compare the average income of the two groups
- The test is called an **independent samples t-test**

"Independent" means the two groups are separate - no household is in both groups. Rural and urban are two different sets of households.

## Test 1: The Code

```
rural = df[df['region_type'] == 'Rural']['income']
urban = df[df['region_type'] == 'Urban']['income']

t, p = stats.ttest_ind(rural, urban, equal_var=False)

print("t =", round(t, 2))
print("p =", round(p, 4))
```

Line 1 selects income from rural households. Line 2 selects income from urban households. Line 4 runs the test.

## Test 1: Reading the Output

```
t = -3.91  
p = 0.0002
```

- **t** tells us how far apart the two groups are, in standard units
- **p** tells us how likely that gap is by chance

Here,  $p$  is essentially zero - far below 0.05. So we say: urban and rural incomes are significantly different.

## Test 2: Comparing the Same Household Twice

- Sometimes we measure the same household twice
- For example, wellbeing before and after a program
- The two measurements are linked - same household
- We use a **paired t-test**
- This test accounts for the link between the two measurements

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A paired test is more accurate when the same people are measured twice. It removes the noise that comes from comparing different households.

## Test 2: The Code

```
# Suppose we have before and after wellbeing scores
before = df['wellbeing_score'].dropna().iloc[:50]
after = before + 2 + (before * 0.01)

t, p = stats.ttest_rel(after, before)

print("t =", round(t, 2))
print("p =", round(p, 4))
```

### Household Twice

`ttest_rel` runs a paired t-test. "Rel" stands for "related" - the two measurements belong to the same household.

## Test 2: Reading the Output

```
t = 45.12  
p = 0.0000
```

- A large **t** (positive or negative) means the change is consistent
- A small **p** means the change is unlikely to be due to chance

Here **p** is essentially zero. The "after" scores are significantly higher than the "before" scores.

## Test 3: Comparing More Than Two Groups

- The t-test compares two groups
- What if we have three or more groups?
- For example, three age groups: 30 – 40, 41 – 50 and 51 – 60
- We use **ANOVA** - Analysis of Variance
- ANOVA checks whether any of the groups differ from the others

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- ANOVA checks whether any of the groups differ from the others

ANOVA is like a t-test that works with any number of groups. It gives us one p-value for the whole comparison.

## Test 3: The Code

```
# Group income by region type
groups = [
    df_thirties['wellbeing_score'].dropna(),
    df_fourties['wellbeing_score'].dropna(),
    df_fifties['wellbeing_score'].dropna()
]

F, p = stats.f_oneway(*groups)

print("F =", round(F, 2))
print("p =", round(p, 4))
```

With only two regions, ANOVA gives the same answer as the t-test. With three or more, it becomes essential.

## Test 3: Reading the Output

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- A large **F** means the groups are more different than random
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Same reading as before:  $p < 0.05$  means the groups differ significantly.

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Correlation does not mean causation. Two variables can move together for many reasons.

## Test 4: The Code

```
r, p = stats.pearsonr(df['education_years'].dropna(),  
                      df['income'].dropna())  
  
print("correlation =", round(r, 2))  
print("p-value      =", round(p, 4))
```

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                      df['income'].dropna())  
  
print("correlation =", round(r, 2))  
print("p-value      =", round(p, 4))
```

pearsonr measures linear correlation. The "r" value tells us how strong the link is. The "p" value tells us how likely the link is by chance.

## Test 4: Reading the Output

```
correlation = 0.28  
p-value     = 0.0001
```

- The correlation is 0.28 - a moderate positive link
- Households with more education tend to have higher income
- The p-value is essentially zero - this is not chance

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A correlation of 0.28 is not perfect. Some households break the pattern. That is normal in social data.

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- Regression gives us a slope: how much income changes when education goes up by one year
- It also gives us a p-value for that slope

Regression turns a relationship into a sentence: "on average, one more year of education adds X to monthly income."

## Test 5: The Code

```
import statsmodels.api as sm

X = sm.add_constant(df['education_years'])
y = df['income']

model = sm.OLS(y, X, missing='drop').fit()

print("Slope:", round(model.params['education_years'], 2))
print("p-value:", round(model.pvalues['education_years'], 4))
```

OLS stands for "Ordinary Least Squares" - a fancy name for the simplest regression. Do not worry about the name.

## Test 5: Reading the Output

```
Slope: 2013.4  
p-value: 0.0001
```

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- The slope is about 2013.4.
- On average, each additional year of education is linked to increase of 2013 more monthly income
- The p-value is essentially zero - the slope is significant

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A significant p-value only means: "this is unlikely to be chance." It says nothing about whether the finding matters.

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Every one of these tests follows the same pattern: run the test, read the p-value, interpret carefully.

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- 5 Regression predicts one variable from another
- 6 Correlation is not causation

**Next: what if the data is not normal?**